## Equation of a tangent from a graph

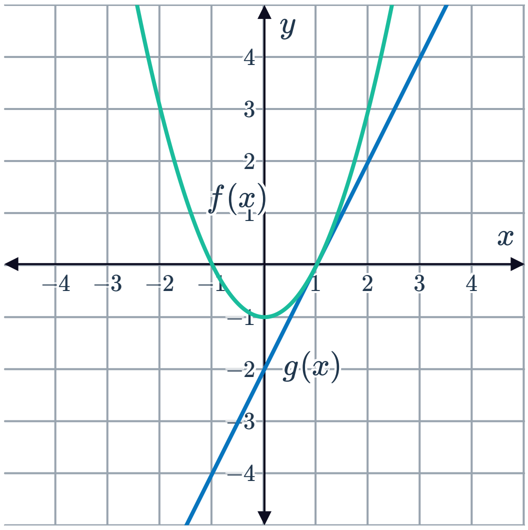
1 Each of the following graphs contain a curve, $f(x)$, along with one of its tangents, $g(x)$.

i State the coordinates of the point at which $g(x)$ is a tangent to the curve $f(x)$.

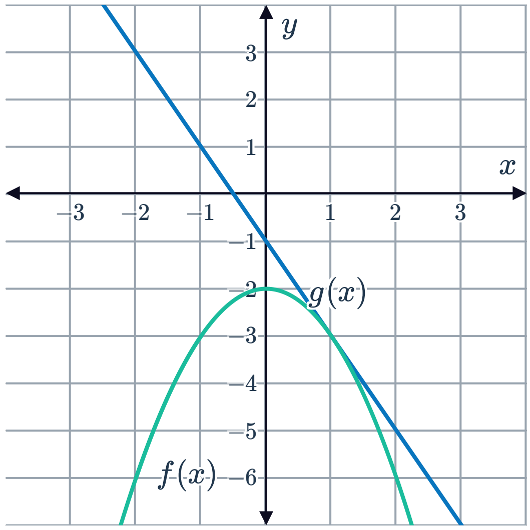
ii State the gradient of the tangent.

iii Hence determine the equation of the line $y = g(x)$.

a




b

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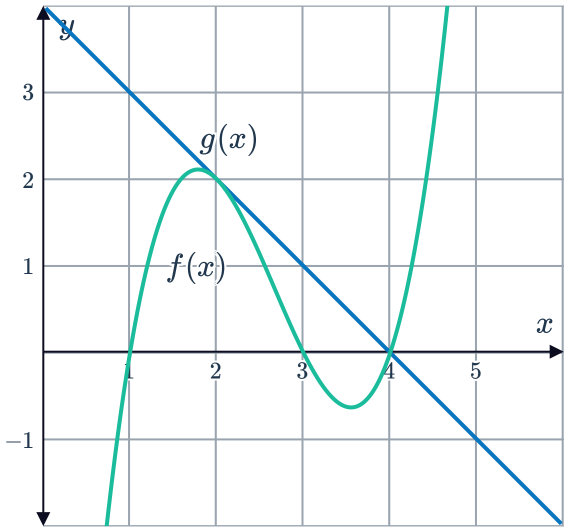


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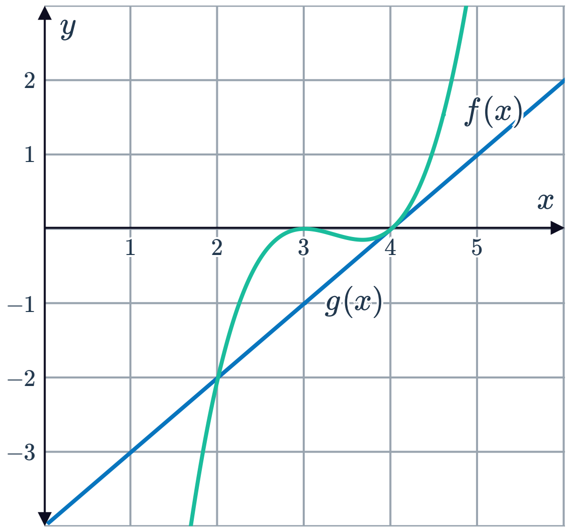
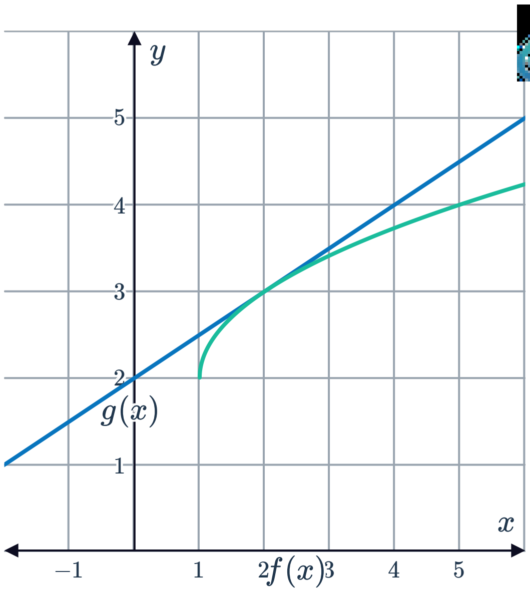


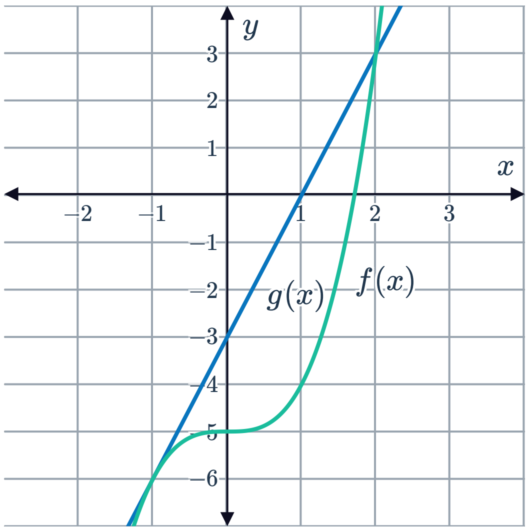
**e**



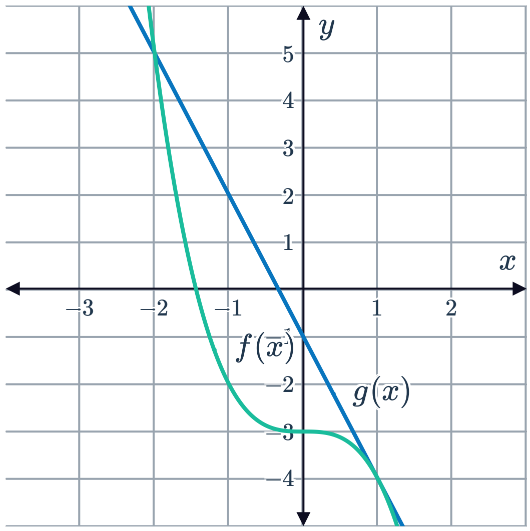

**2** Each of the following graphs contain a curve, $f(x)$, along with one of its tangents, $g(x)$.

- **i** State the coordinates of the point at which $g(x)$ is a tangent to the curve $f(x)$.
- **ii** State the gradient of the tangent.
- **iii** Hence determine the equation of the line $y = g(x)$.
- **iv** State the $x$-coordinate of the point on the curve at which we could draw a tangent that has the same gradient as $g(x)$.

**a**



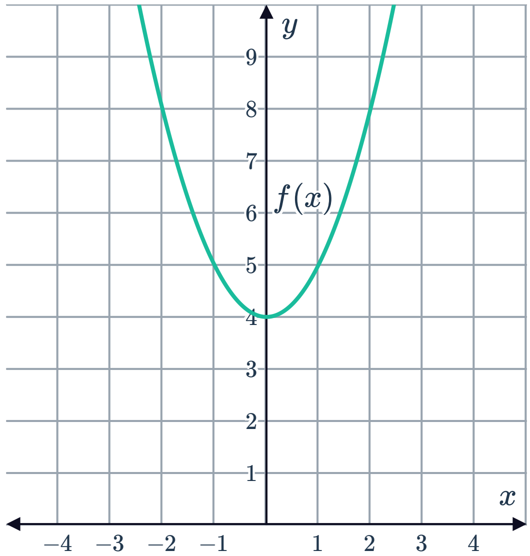

**b**



**3** Consider the graph of the function $f(x)$:

**a** Sketch the graph of the function $g(x) = 2x + 3$ on the same number plane.

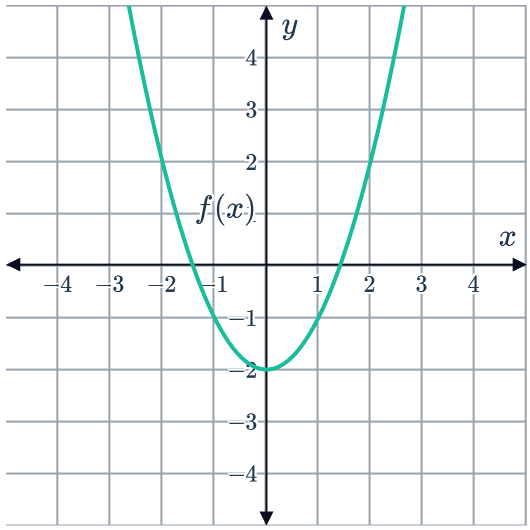
**b** Is $g(x)$ a tangent to $f(x)$? Explain your answer.




**4** Consider the graph of the function $f(x)$:

**a** Sketch the graph of the function $g(x) = 2x - 1$ on the same number plane.

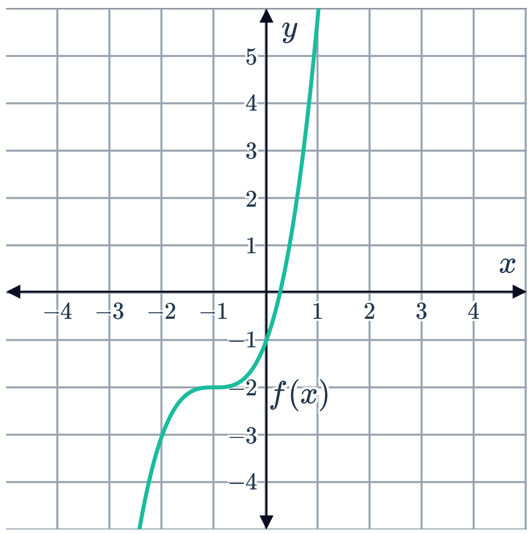
**b** Is $g(x)$ a tangent to $f(x)$? Explain your answer.



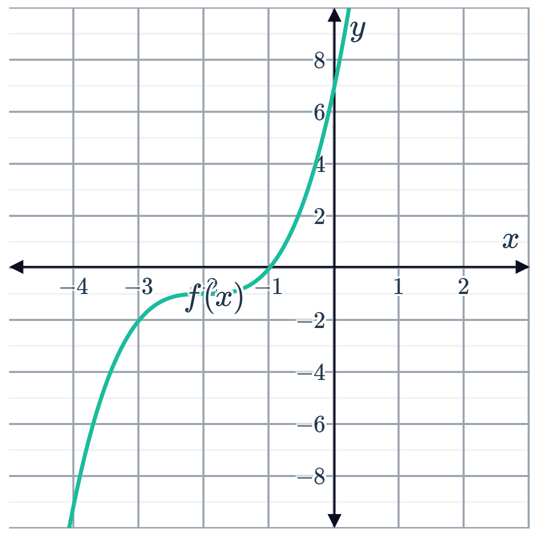

**5** Consider the graph of the function $f(x)$:

**a** Sketch the graph of the function $g(x) = 3x + 3$ on the same number plane.

**b** Is $g(x)$ a tangent to $f(x)$? Explain your answer.



6 Consider the graph of the function $f(x)$:




a Sketch the graph of the function $g(x) = 4x + 7$ on the same number plane.

b Is $g(x)$ a tangent to $f(x)$? Explain your answer.

7 Consider the curve given by the function $f(x) = x^2 - 1$.

a Find the gradient of the tangent to the curve at the point $(1, 0)$.

b Graph the curve and the tangent at the point $(1, 0)$ on a number plane.

8 Consider the curve given by the function $f(x) = x^2 - 4x + 2$.

a Find the gradient of the tangent to the curve at the point $(3, -1)$.

b State the coordinates of the vertex of the parabola $f(x) = x^2 - 4x + 2$.

c Graph the curve and the tangent at the point $(3, -1)$ on a number plane.

## Equation of a tangent at a given point

9 Consider the function $y = x^2 - 3x + 4$. State the $x$-coordinate of the point on the curve where the tangent makes an angle of $45^\circ$ with the $x$-axis.

10 Consider the tangent to the curve $f(x) = x^2$ at the point $(-1, 1)$.

a Find the gradient of the function $f(x) = x^2$ at this point.

b Find the equation of the tangent to the curve $f(x) = x^2$ at the point $(-1, 1)$.

11 Consider the tangent to the curve $f(x) = x^3$ at the point $(-1, -1)$.

a Find the gradient of the function $f(x) = x^3$ at this point.

b Hence find the equation of the tangent to the curve $f(x) = x^3$ at the point $(-1, -1)$.

**12** Consider the tangent to the curve $f(x) =$ [redacted]t the point $(-3, -9)$.

a Find the gradient of the function $f(x) = -x^2$ at this point.

b Hence find the equation of the tangent to the curve $f(x) = -x^2$ at the point $(-3, -9)$.

**13** Consider the tangent to the curve $f(x) = x^2$ at the point $(-2, 4)$.

a Find the gradient of the function $f(x) = x^2$ at this point.

b Find the equation of the tangent to the curve $f(x) = x^2$ at the point $(-2, 4)$

**14** Consider the function $f(x) = 3x^2$.

a Find the gradient of the function at $x = 2$

b Find the $y$-coordinate of the point on the function at $x = 2$.

c Hence find the equation of the tangent to the curve $f(x) = 3x^2$ at $x = 2$.

**15** Find the equation of the tangent to the curve $f(x) = 0.3x^3 - 5x^2 - x + 4$ at $x = 1$.

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## Applications

**16** For each of the following:

i Find the $x$-coordinate of point $M$.

ii Find the $y$-coordinate of point $M$.

a At point $M(x, y)$, the equation of the tangent to the curve $y = x^2$ is given by $y = 4x - 4$.

b At point $M(x, y)$, the equation of the tangent to the curve $y = x^3$ is given by $y = 12x - 16$.

**17** Consider the function $f(x) = \dfrac{4x^3}{3} + \dfrac{5x^2}{2} - 3x + 7$. Find the $x$-coordinates of the points on the curve whose tangent is parallel to the line $y = 3x + 7$.

**18** $5x + y + 2 = 0$ is the tangent to the curve $y = x^2 + bx + c$ at the point $(9, -47)$.

a Find the derivative $\dfrac{dy}{dx}$ of $y = x^2 + bx + c$.

b Find the gradient of the tangent to the curve at $x = 9$.

c Solve for the value of $b$.

d Solve for the value of $c$.

**19** Consider the function $f(x) = x^2 + 5x$.

- **a** Find the $x$-coordinate of the point at which $f(x)$ has a gradient of $13$.
- **b** Hence state the coordinates of the point on the curve where the gradient is $13$.

**20** Consider the function $f(x) = x^3 - 6x^2$.

- **a** Find the $x$-coordinates of the points at which $f(x)$ has a gradient of $495$.
- **b** Hence state the coordinates of the points on the curve where the gradient is $495$.

**21** Consider the function $f(x) = x^3 + 3x^2 - 19x + 2$.

- **a** Find the $x$-coordinates of the points at which $f(x)$ has a gradient of $5$.
- **b** Hence state the coordinates of the points on the curve where the gradient is $5$.

**22** Consider the function $f(x) = x^3 + 6x^2 - 14x - 2$.

- **a** Find the $x$-coordinates of the points on the curve where the gradient is the same as that of $g(x) = 22x - 4$.
- **b** Hence state the coordinates of the points on the curve where the gradient is the same as that of $g(x) = 22x - 4$.

**23** The curve $f(x) = k\sqrt{x} - 5x$ has a gradient of $0$ at $x = 16$.Find the value of $k$.

**24** Consider the function $y = 4x^2 - 5x + 2$.

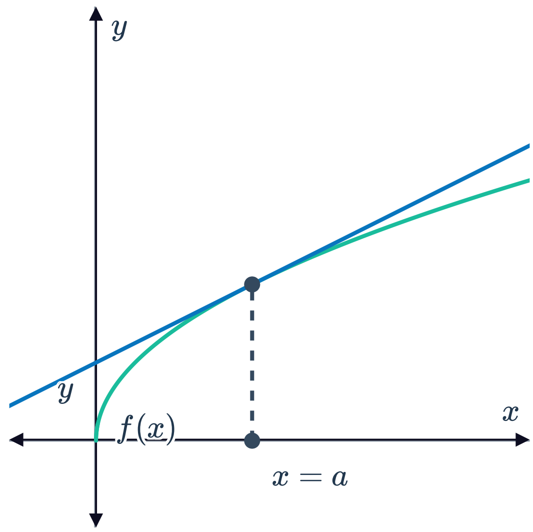
- **a** Find $\frac{dy}{dx}$.
- **b** Hence find the value of $x$ at which the tangent to the parabola is parallel to the $x$-axis.

**25** Consider the function $f(x) = 5x^2 + \frac{4}{x} - 1$. The tangent to the curve at the point $(2, 21)$ makes an angle of $\theta$ with the $x$-axis. Find $\theta$, correct to the nearest degree.

**26** The curve $y = ax^3 + bx^2 + 2x - 17$ has a gradient of $58$ at the point $(2, 31)$. Find the values of $a$ and $b$.

**27** The graph of $y = ax^3 + bx^2 + cx + d$ intersects the $x$-axis at $(2, 0)$, where it has a gradient of $36$. It also intersects the $y$-axis at $y = -28$, where the tangent is parallel to the $x$-axis. Find the values of $a, b, c$ and $d$.

**28** In the following graph, the line $y = \dfrac{x}{10} + b$ is a tangent to the graph of $f(x) = 6\sqrt{x}$ at $x = a$.




**a** Find the value of $a$.

**b** Find the value of $b$.